Wiring & Assembly Guide: Gruner 740 Dual-Coil Relay with Raspberry Pi 5

This reference document provides exact, step-by-step instructions for assembling your bench-test validation proof of concept. Do not apply live mains voltage (240V) during this initial testing phase.

1. Hardware Pin Mapping Reference

Raspberry Pi 5 Pin	SunFounder Pin	Wire Function
Pin 2 (5V Power)	VCC	Powers board logic & onboard relay coils
Pin 6 (GND)	GND	Establishes a common signal ground
Pin 11 (GPIO 17)	IN1	Controls Relay 1 (Throws Contacts CLOSED)
Pin 13 (GPIO 27)	IN2	Controls Relay 2 (Throws Contacts OPEN)

2. Step-by-Step Bench Assembly Sequence

Step A: Ensure your Raspberry Pi 5 is powered off and the ALITOVE 12V DC power adapter is unplugged from the wall outlet.

Step B: Locate the small black plastic jumper cap on the SunFounder board bridging *VCC* and *JD-VCC*. **Leave this jumper completely intact.**

Step C: Use four of your ELEGOO Female-to-Female jumper wires to connect the low-voltage control pins between the Raspberry Pi 5 header and the SunFounder module input strip exactly as detailed in the Pin Mapping table above.

Step D: Wire the high-current 12V coil loop. Take the **Black wire (Common Ground)** from the SunPower Gruner relay and run it straight to the **Negative (-) screw terminal** of your ALITOVE 12V power supply terminal adapter.

Step E: Take the **Positive (+) wire** from your ALITOVE 12V power adapter and split or chain it into the **Center Common (COM)** screw terminals of both Relay 1 and Relay 2 on the SunFounder board.

Step F: Connect the Gruner relay's **Red wire (Close Coil)** into the **Normally Open (NO)** screw terminal of SunFounder Relay Channel 1.

Step G: Connect the Gruner relay's **Green wire (Open Coil)** into the **Normally Open (NO)** screw terminal of SunFounder Relay Channel 2.

Step H: Clamp two short, unpowered 12-inch pieces of your 3/0 AWG conductor lines inside the heavy mechanical lugs of the Gruner relay to act as visual position anchors during bench testing.

3. Python Control Script Implementation

```
# Save as gruner_control.py on your Raspberry Pi 5
import time
from gpiozero import DigitalOutputDevice

# Low-level triggers: active_high=False fires relay when GPIO goes low
close_switch = DigitalOutputDevice(17, active_high=False, initial_value=False)
open_switch = DigitalOutputDevice(27, active_high=False, initial_value=False)

PULSE_DURATION = 0.05  # 50 milliseconds (Meets Gruner's >=30ms minimum)

def pulse_close():
    print("FIRING: Activating Red Wire (Closing 200A contacts)...")
    close_switch.on()
    time.sleep(PULSE_DURATION)
    close_switch.off()
    print("Coil completely isolated.")

def pulse_open():
    print("FIRING: Activating Green Wire (Opening 200A contacts)...")
    open_switch.on()
    time.sleep(PULSE_DURATION)
    open_switch.off()
    print("Coil completely isolated.")

try:
    pulse_close()
    time.sleep(3)
    pulse_open()
finally:
    close_switch.off()
    open_switch.off()
```